

Use Case Process Report : Syensqo

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Context of the Use-Case

Syensqo, a Belgian company in fine chemistry originating from the Solvay group's split in 2023, identifies the health and safety of its 13,000 employees as its absolute priority, particularly within its global laboratory network.

Since 2019, the company has accumulated an **extensive database of over 12,000 reporting entries** of incidents which happened on their sites documented in **natural language**. Despite this wealth of information, the massive volume of data makes it increasingly difficult for teams to manually extract actionable insights or identify emerging risk trends.

To address this, the current project aims to develop a **prescriptive AI tool** that deciphers this natural language corpus and **integrate it to an API** to provide more efficiency in decision support for HSE managers, helping them treat hazardous situations and **implement preventive measures to reduce workplace accidents**.

Objectives of the product

Our objective is to develop a **prescriptive decision-support tool** that transforms Syensqo's database of more than 12,000 records into actionable safety insights. By using LLMs to process the natural language parts found in incident reports, the tool identifies risk trends that might otherwise go unnoticed as HSE managers don't have time to read and concatenate every report's description. This approach allows us to propose **preventive recommendations** to **minimize workplace accidents and injuries**, exactly as commissioned by the company.

POC

To reach these goals, we have thought of a **decision-support portal** designed to condense the 12,000 entries of the database for managers to use. This dashboard will **display graphs** presenting the repartition of reports by multi-dimensional filtering as well as an **Ollama-based chatbot** by RAG on the database to highlight high-risk patterns, prevent security breaches and provide short-term solutions for new incidents.

Before integrating the data into the portal, we implemented a processing pipeline to ensure the 12,000+ entries were optimized for the LLM and visualization tools :

- **Format Conversion:** We processed the source Parquet files using Python (Pandas) to **ensure high-performance reading** and compatibility with our local environment.
- **Text Pre-processing:** We focused on the "Description", "Actions taken" and "Recommandations" fields, **removing redundant data** to clear the path for our local LLM.
- **Categorization Alignment:** We maintained the strict distinction between Concerns, Near Misses, and Injuries to allow the chatbot to **prioritize lessons learned from actual accidents** versus potential risks.
- **Vectorization for RAG:** The cleaned natural language descriptions were converted into vector embeddings. These are stored in a local vector database, allowing the Ollama-based chatbot to **perform semantic searches** and retrieve relevant precedents.

Solution development

To work more efficiently, we divided our group in 3 smaller group :

- **Analytics & Visualization (4 people):** This group focused on data cleaning and the design of the HSE Portal. They developed an **interactive dashboard** that translates the 12,000+ entries into **multi-dimensional graphs**, allowing managers to filter by site, date, and incident type.
- **Data Engineering & Clustering (2 people):** This team focused on preparing the unstructured text for the AI. By **vectorizing the "Description" field**, they identified latent patterns and clusters in the reports, providing a structured knowledge base for the RAG system to navigate.
- **AI & RAG Implementation (5 people):** This group developed the core intelligence of the tool using an Ollama-based LLM. They implemented a **Retrieval-Augmented Generation (RAG)** pipeline to ground the chatbot in past reports and a **SQL request system**. This allows the AI to perform real-time statistical analysis and provide prescriptive recommendations based on past "Actions taken."

MVP

This project delivers a **Minimum Viable Product (MVP)** designed to validate the feasibility in a single week and business value of a dual-approach solution: an interactive visual dashboard combined with an intelligent conversational assistant.

1. The Interactive Dashboard

The first component is a functional dashboard designed to provide managers with immediate situational awareness. It visualizes key metrics, such as the frequency and typology of incidents across different work sites. The interface allows for dynamic interaction; users can filter views, click on controls, and hover over charts to explore the underlying data deeply and intuitively.

2. Operation of the AI Pipeline

The chatbot operates on a hybrid AI pipeline designed to analyze user queries and provide reliable, contextualized HSE responses.

1. User Query Analysis and Routing

When a user submits a question, it is first processed by a **routing and optimization LLM (Large Language Model)**.

This model has two main objectives:

- determine whether the question requires a **SQL query** (quantitative data, counts, aggregations)
- reformulate and optimize the query to improve both **semantic search** and **SQL generation**

2. SQL Query Generation and Execution (if required)

If the routing LLM determines that a SQL query is necessary:

- the optimized query is sent to a **Text-to-SQL specialized LLM**, trained to generate correct and secure SQL queries

the generated SQL query is executed on the HSE database

- the **SQL results**, considered the authoritative source for quantitative data, are retrieved

3. Semantic Retrieval via RAG (Retrieval-Augmented Generation)

In parallel (or exclusively when no SQL query is required), the optimized query is processed by the **RAG pipeline**:

- embeddings are used to perform **similarity-based semantic search**
- the **12 most relevant incidents** are retrieved from the vector database
- these excerpts provide **factual context** based on real historical incidents

4. Context Construction for the Final LLM

The following elements are then combined:

the system prompt, which strictly defines the HSE role of the assistant

the SQL results (when available)

the 12 most similar incidents retrieved via RAG

the optimized query

This enriched context is passed to the **Final LLM**, ensuring that the response is grounded in concrete data and aligned with HSE objectives.

5. Response Generation and Display

The **Final LLM** generates a clear response focused on prevention and managerial decision-making.

The response is then **displayed to the user through the Streamlit interface**.

Conclusion

In conclusion, we have developed a functional tool tailored to the specific HSE management challenges faced by Syensqo.

By implementing data clustering, vectorization, and an **Ollama-based chatbot grounded via RAG**, we have transformed the complex issue of unstructured natural language reports into a proactive safety asset. This system, integrated into a multi-dimensional dashboard, provides a comprehensive overview of global site statistics while enabling managers to identify precursors to accidents.

Ultimately, this tool gives HSE managers an opportunity to shift from reactive monitoring to **predictive prevention**, ensuring the physical integrity of collaborators remains the group's top priority.